

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5_AT3 — INTERACTIVE DASHBOARD PROJECT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Create (BL6)	Assessment:	Assessment Tool 3 – Interactive Dashboard Project
Weightage:	20% of CIA	Total Marks:	25 (Dataset: 7, Interactivity: 9, Effective Visualization: 9)
CO5 Statement:	<i>Create an original, interactive dashboard that selects and prepares an appropriate dataset, implements meaningful software interactivity, and applies effective visualization design principles.(BL6)</i>		
Student Name:	Kola Amrutha Sandhya	Reg. No.:	192524270
Section / Batch:	AIDS	Date:	29-08-2026

Code of Conduct

I Kola Amrutha Sandhya 192524270 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Amrutha

Instructions

- This is an individual project. Each student designs, builds, and submits one original interactive dashboard. Total: 25 Marks.
- Electronic devices and software tools are expected and permitted — use any dashboarding tool (e.g., Tableau, Power BI, Plotly Dash, Streamlit, or similar) of your choice.
- Late submissions and academic integrity are governed by the standard course policy; all code/data sources must be cited.
- Submission window: as announced by the instructor (project, not a timed test).
- Choose and clearly justify your own dataset — it must be real or realistic, relevant to a stated purpose, and appropriately cleaned/prepared before use.
- The dashboard must include genuine software interactivity (e.g., filters, zoom/pan, hover tooltips, drill-down, or linked views) — a static image or PDF does not satisfy this requirement.
- Apply effective visualization principles covered in this course (proportional ink, appropriate axis/color scale choice, avoiding overplotting, accessibility) throughout the dashboard.
- Submit both the working dashboard (or a recorded walkthrough, if live hosting is not possible) and a short written justification of your dataset and design choices.

Annexure A – INTERACTIVE DASHBOARD PROJECT

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT3 — Interactive Dashboard Project

CO5 · BT6: Create ·

This rubric is used to assess the CO5-AT3 Interactive Dashboard Project, in which each student designs, builds, and submits an original interactive dashboard, evaluated on dataset quality, software interactivity, and effective visualization design.

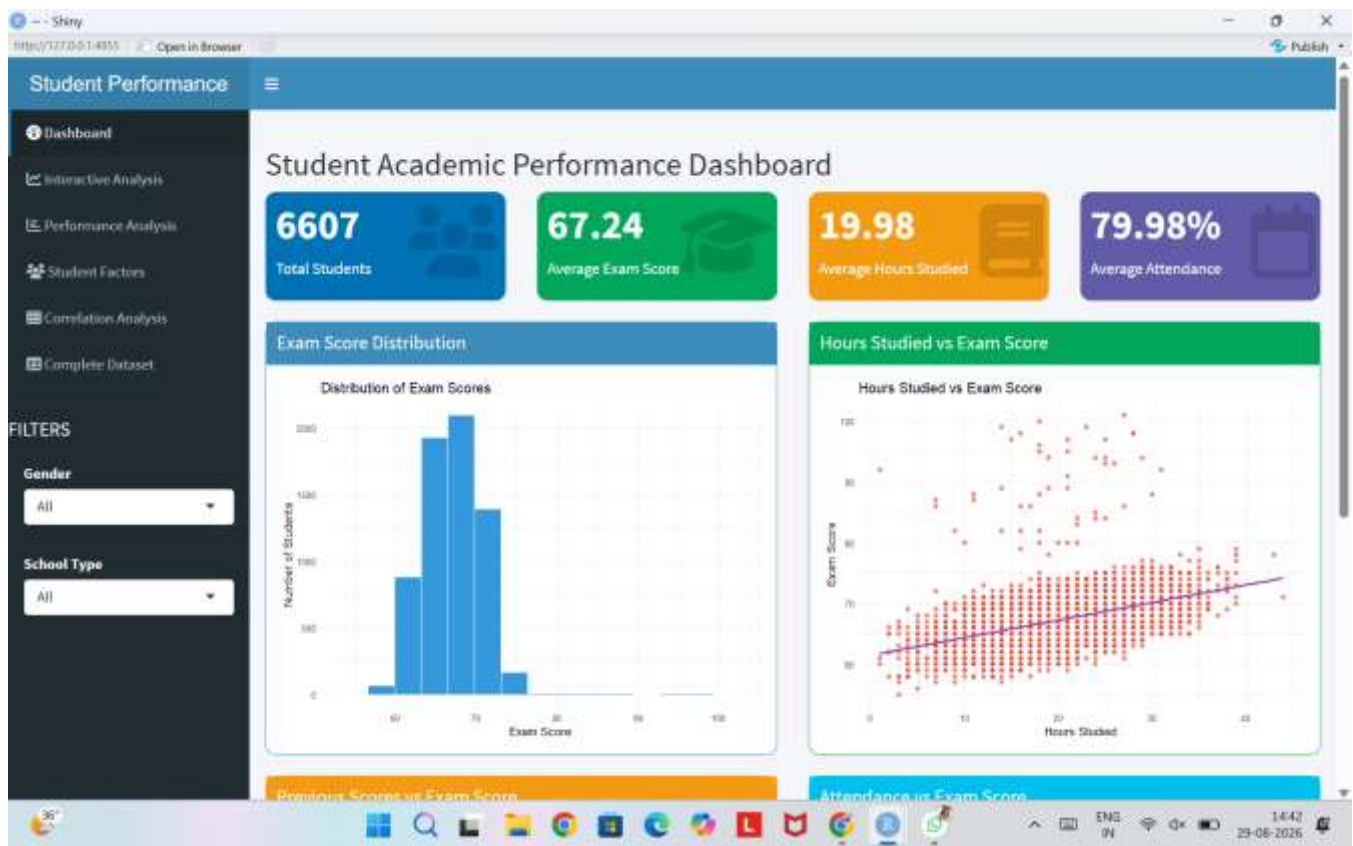
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Dataset (Selection, Quality & Preparation)	30%	Dataset is highly relevant to a clearly stated purpose, real or realistic, appropriately sized, and thoroughly cleaned/prepared with any limitations disclosed.	Dataset is relevant and reasonably prepared, with only minor gaps in cleaning or justification.	Dataset is loosely relevant or only partially prepared; justification is weak or missing.	Dataset is irrelevant, unprepared, or trivial (e.g., a toy dataset with no real justification).
Interactivity of the Software/Dashboard	35%	Dashboard includes multiple well-implemented, meaningful interactive features (e.g., filters, drill-down, linked views,	Dashboard includes working interactive features that function correctly, with only minor gaps in polish or	Dashboard includes limited or partially working interactivity; some features are superficial or broken.	Dashboard is static or interactivity does not function; the submission does not meet the core project requirement.

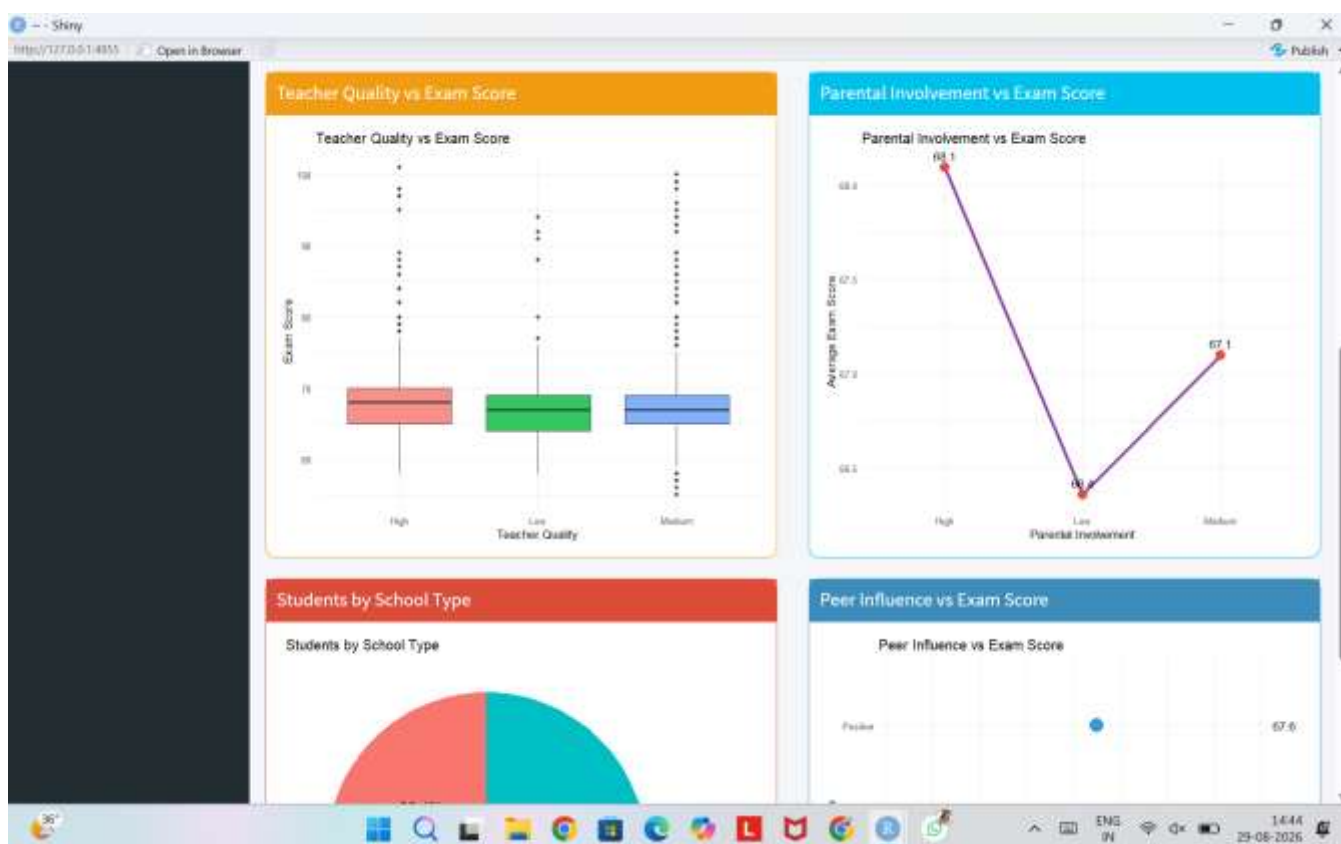
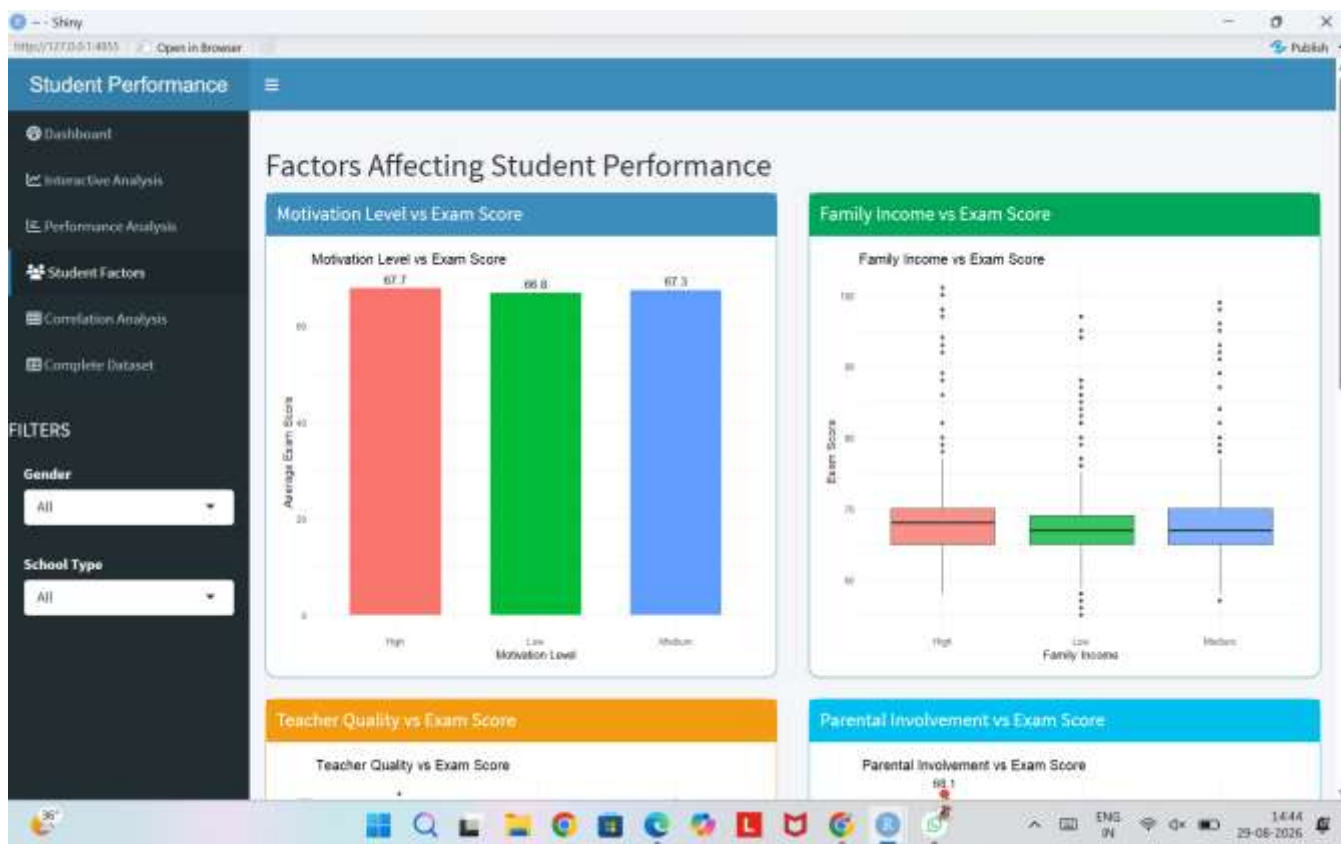
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		tooltips) that genuinely help the user explore the data.	scope.		
Effective Visualization Design	35%	Chart types, color scales, axis choices, and layout are all well-justified and consistently apply the effective-visualization principles covered in this course.	Visualization choices are mostly appropriate and apply course principles, with minor lapses.	Visualization choices show some understanding of course principles but include noticeable design flaws (e.g., misleading axis, poor color scale).	Visualization choices are poor or arbitrary, showing little evidence of applying course principles.

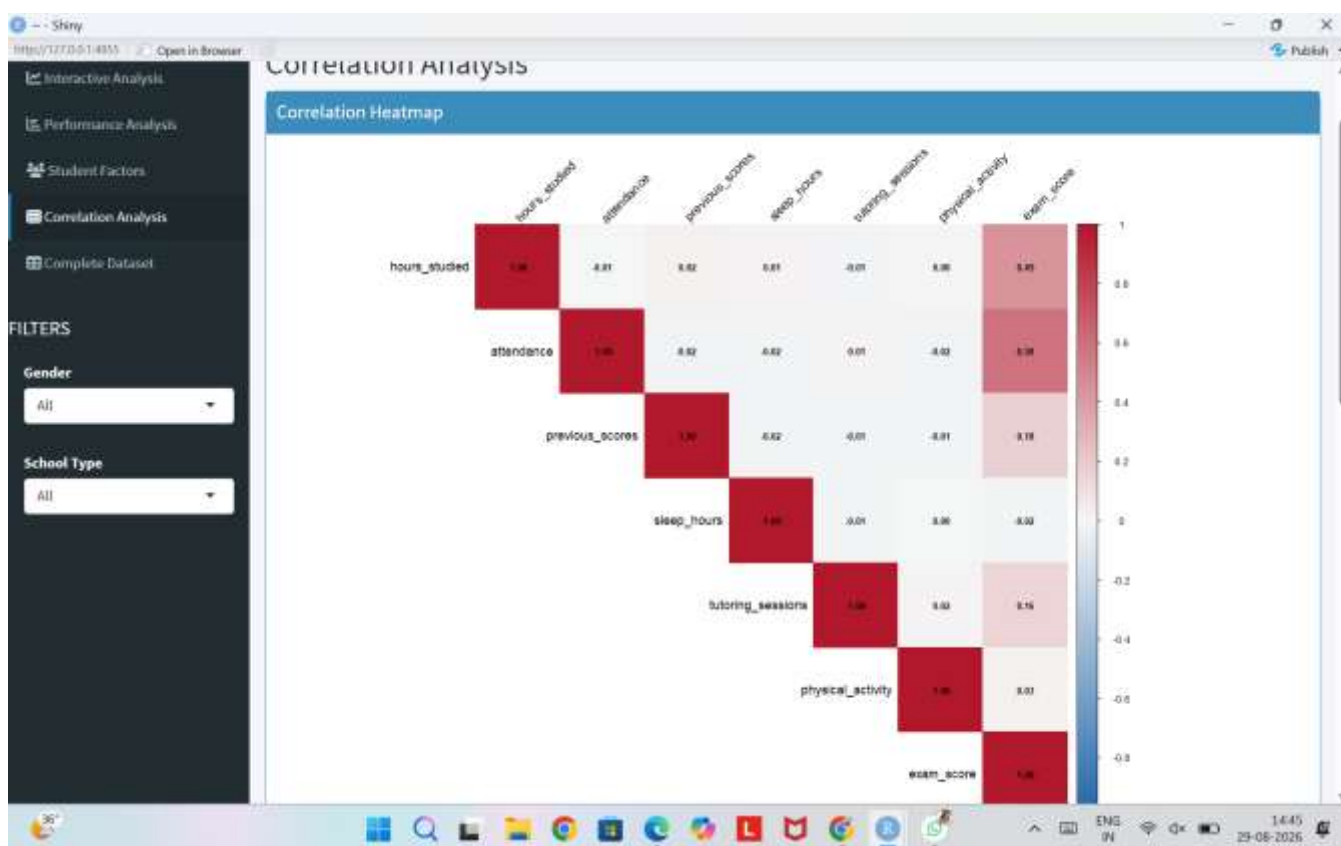
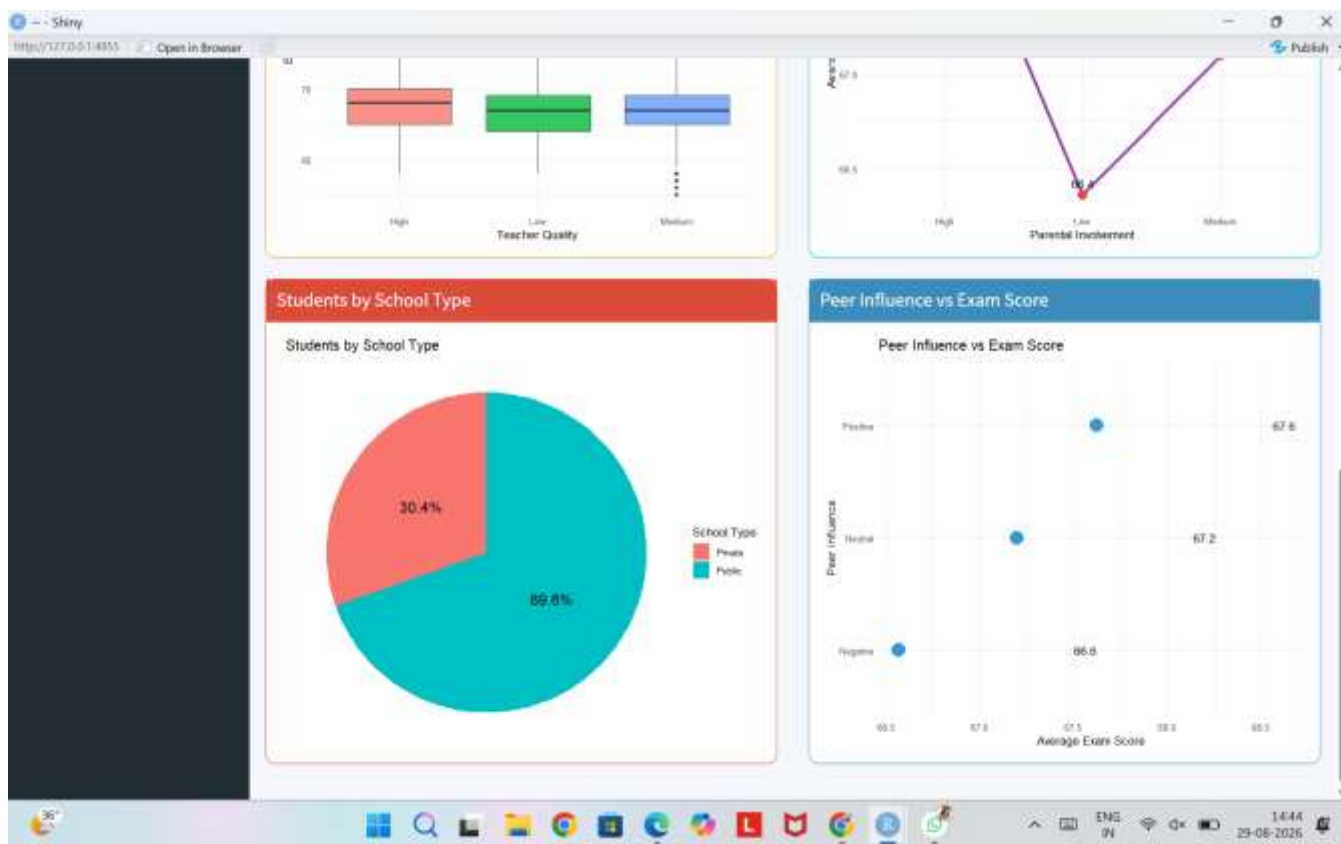
Scoring Guide

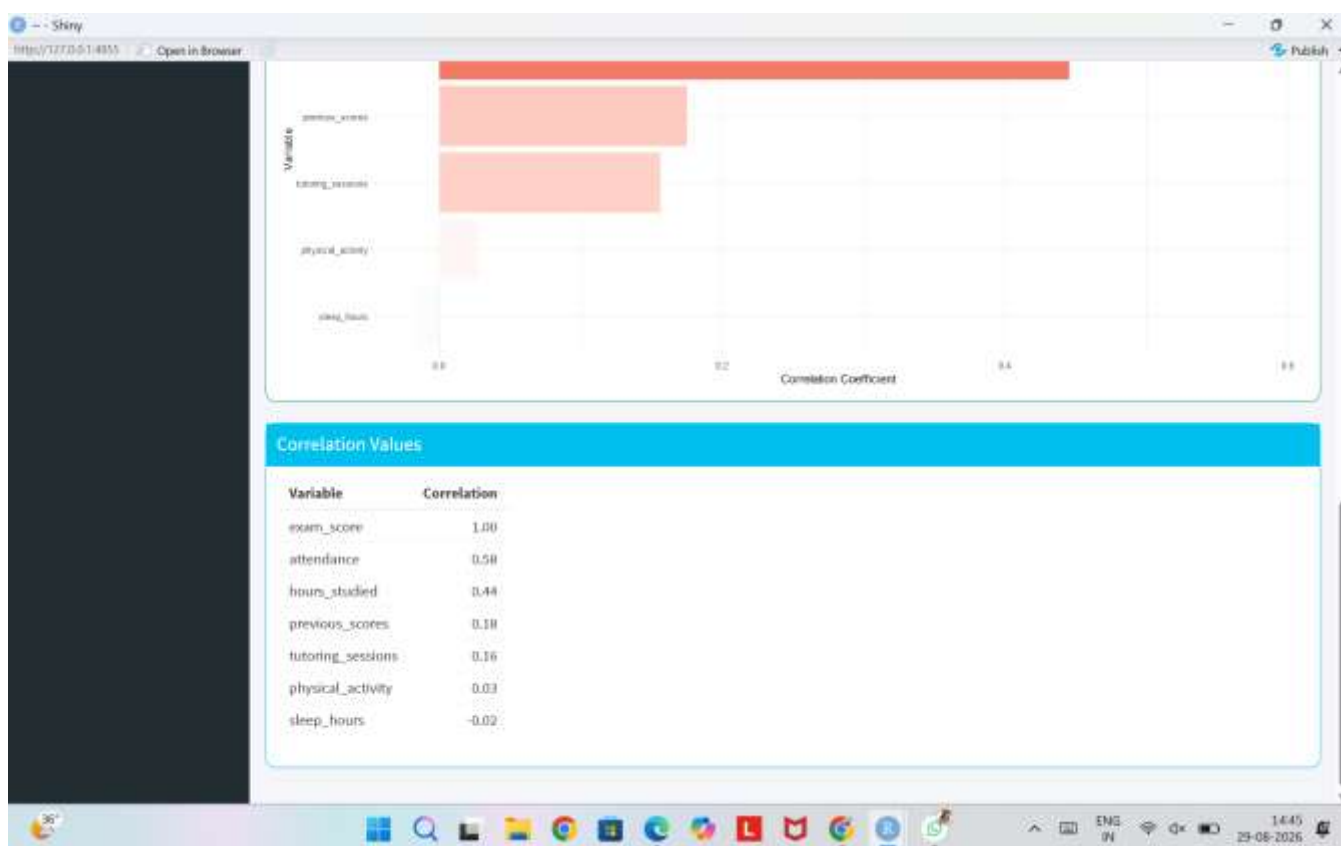
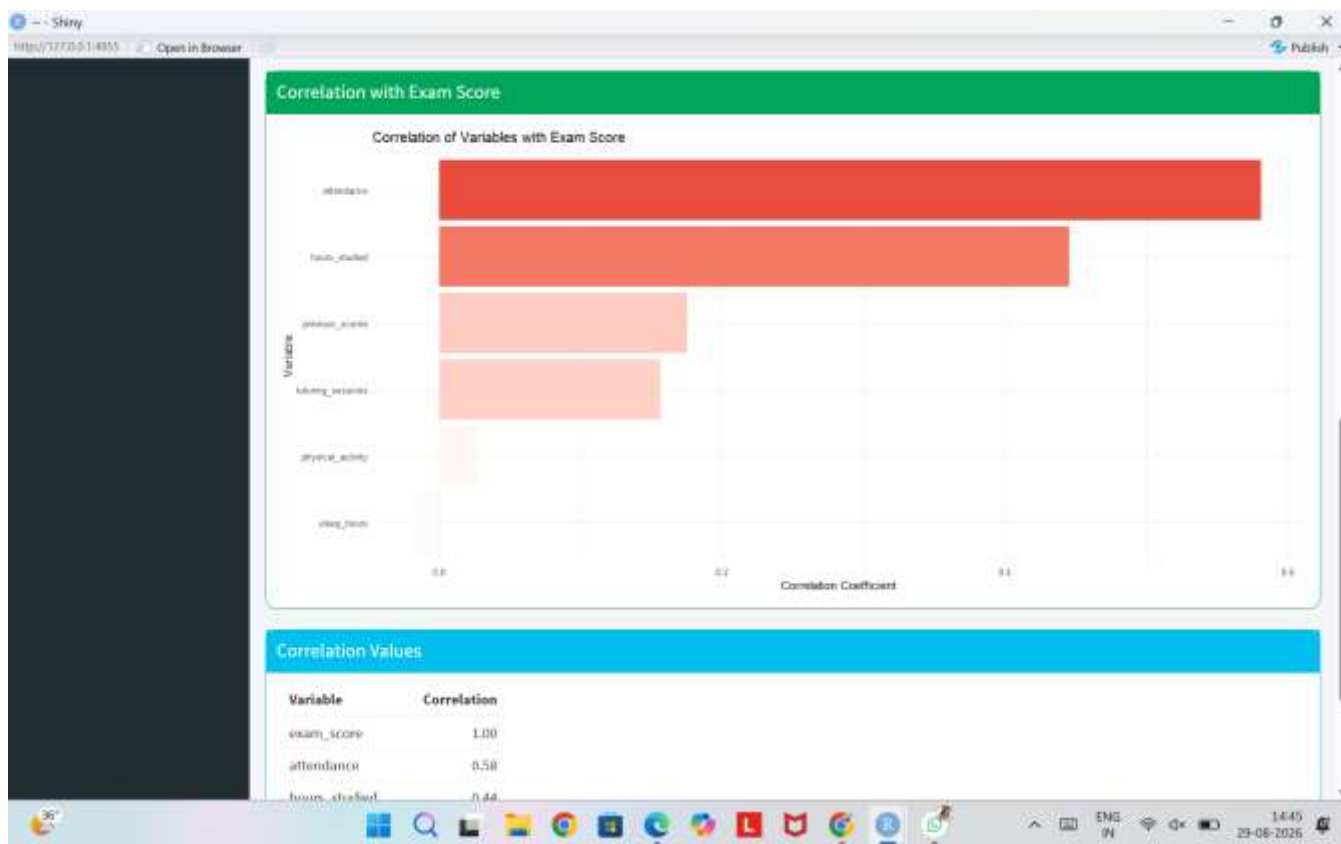
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

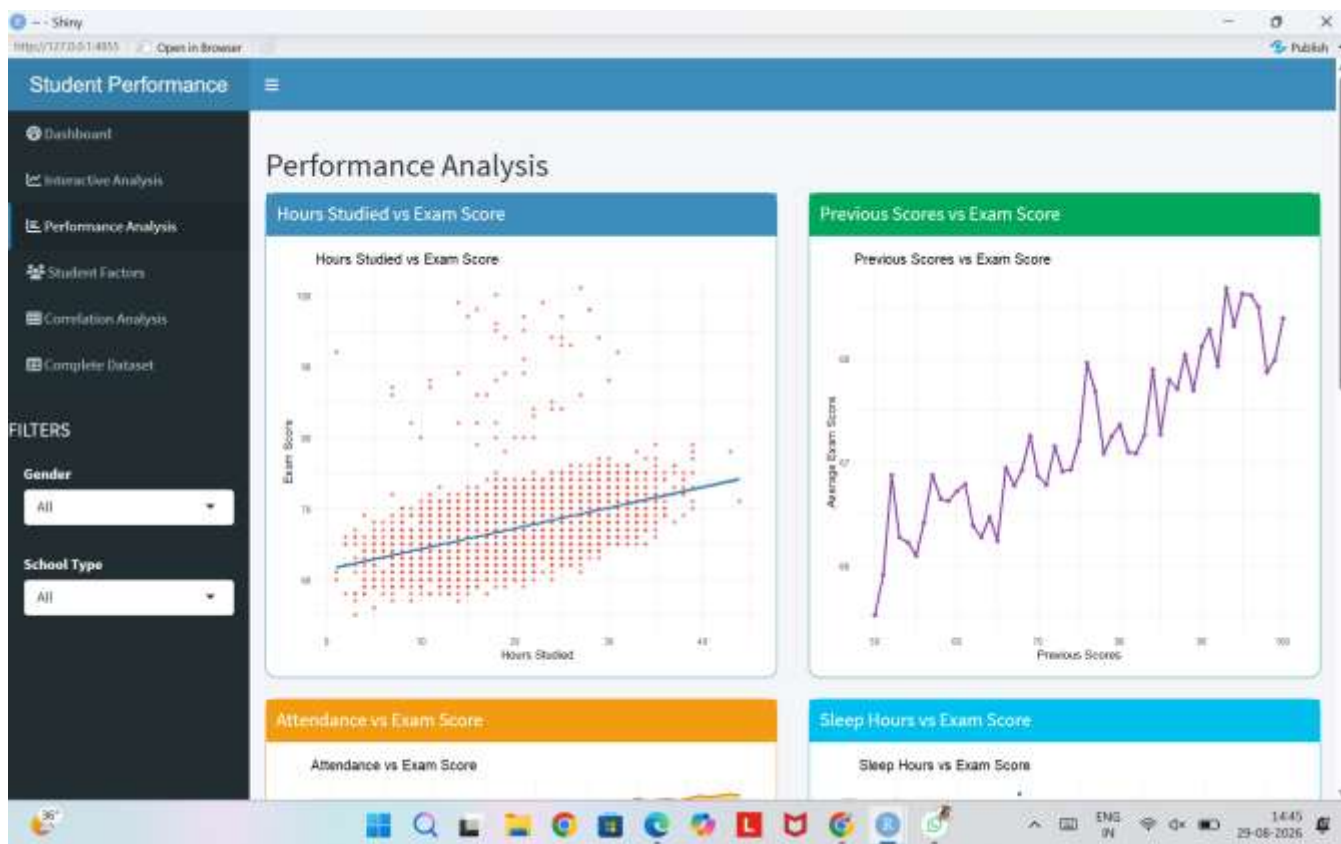
Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

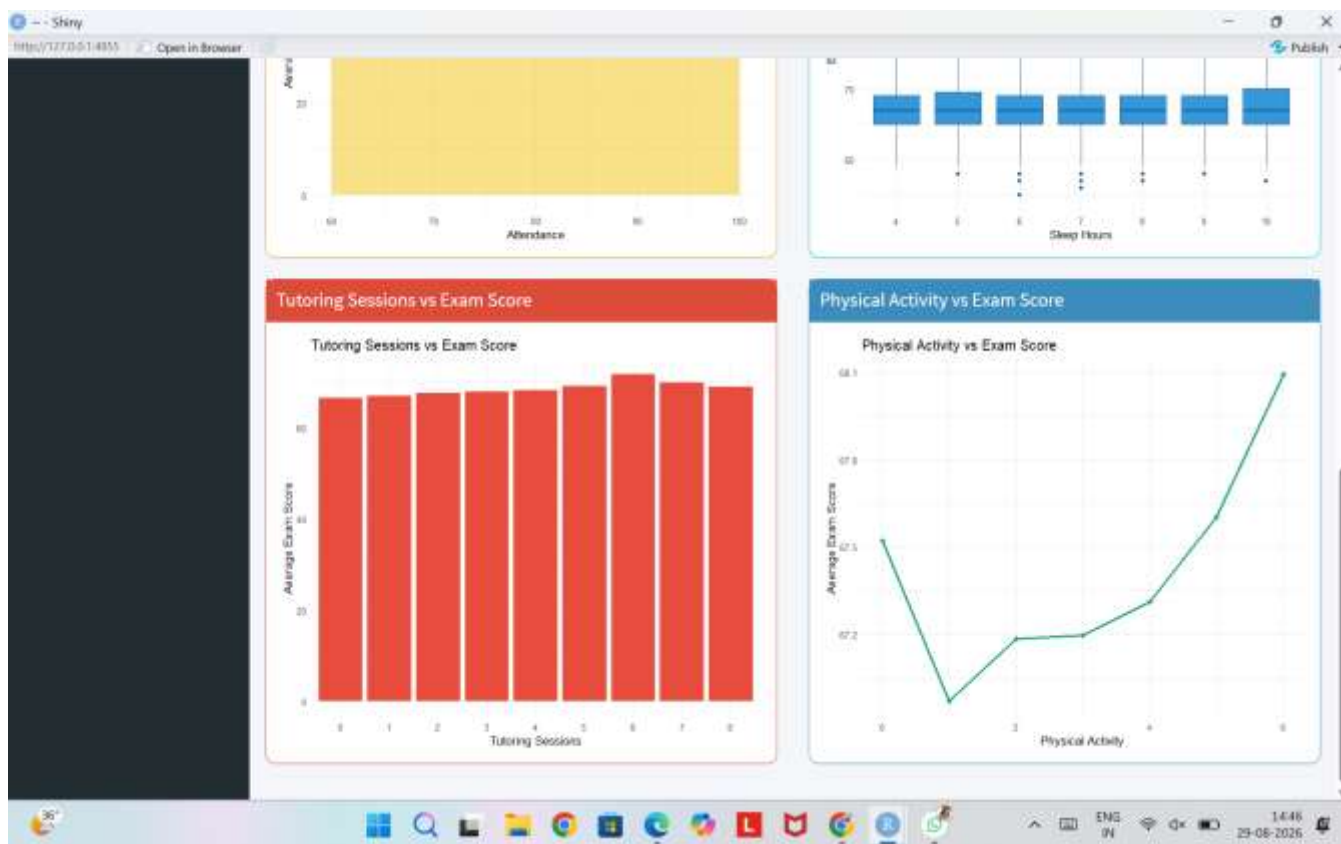












Shiny

https://127.0.0.1:4555 Open in Browser Publish

Student Performance

- Dashboard
- Interactive Analysis
- Performance Analysis
- Student Factors
- Correlation Analysis
- Complete Dataset

Complete Dataset

6607 Total Dataset Rows

20 Total Dataset Columns

6607 Complete Rows

All Dataset Values

Copy CSV Excel Print Search:

hours_studied	attendance	parental_involvement	access_to_resources	extracurricular_activities	sleep_hours	previous_scores
All	All	All	All	All	All	All
23	84	Low	High	No	7	
19	64	Low	Medium	No	8	
24	98	Medium	Medium	Yes	7	
29	89	Low	Medium	Yes	8	
19	92	Medium	Medium	Yes	6	
19	88	Medium	Medium	Yes	8	
29	84	Medium	Low	Yes	7	

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